Electronics lab

FINAL EXAM

2025-Second semester

Student name:___________________ std. no.:________________

Q1: Choose the correct answer – fill the table at the end

1. The unit of I_B in a BJT is:
   a) I mA
   b) I μA
   c) I A
   d) Watt

2. A FET is tested using a DMM, the resistance from gate to source should be:
   a) very Low resistance
   b) Zero
   c) very high resistance
   d) not mentioned

3. In a Half-Wave Rectifier (HWR), the output frequency observed on the oscilloscope is:
   a) Same as input
   b) Half of input
   c) Double the input
   d) Quarter of input

4. In Fixed Bias BJT, the input characteristic curve is:
   a) V_{BE} vs I_B for different V_{CB} values
   b) V_{BE} vs I_B for different V_{CB} values
   c) V_{CE} vs I_C for different I_B values
   d) V_{CE} vs I_C for different V_B values

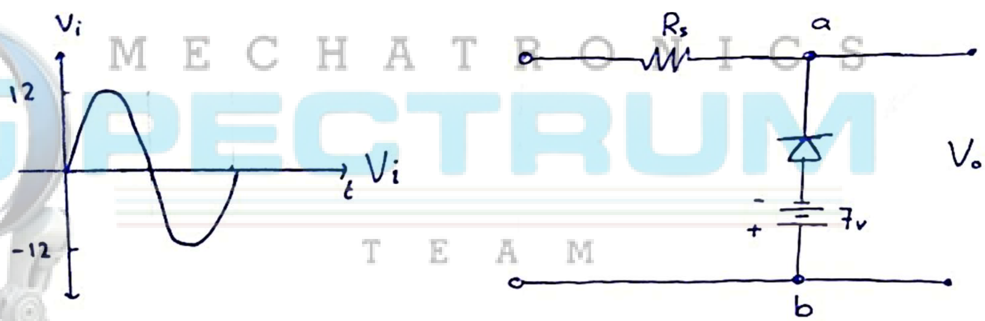
5. In diode characteristics, when the voltage exceeds point A for a normal diode is…….._,and for a zener is ………… :
   a) damaged/working
   b) working/damaged
   c) damaged/damaged
   d) working/working

6. For a clamber circuit, if the output voltage = 15v peak voltage, then the Vrms is:
   a) 15V b) 10.6 c) 5 d) 21.21v

7. In a rectifier circuit, increased ripple means:
a) More effective filtering with higher DC voltage
b) More effective filtering with less DC voltage
c) less effective filtering with higher DC voltage
d) less effective filtering with lower DC voltage

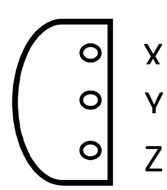
8. While using a fully functional DMM to test a BJT, with the positive to B and negative to C, the reading will be:
a) 0v
b) O.L (over load)
c) around 0.7v
d) not determined

9. Find the voltage across point a,b For the positive half cycle, not that the diode is a silicone diode
a) 7V
b) +12V
c) 0V
d) -12V



10. For the same circuit, If the Resistance = 2kΩ and the diode is ideal, then the current through the resistor equals:
a) 0A
b) 0.35 mA
c) 1 mA
d) 2.5 mA

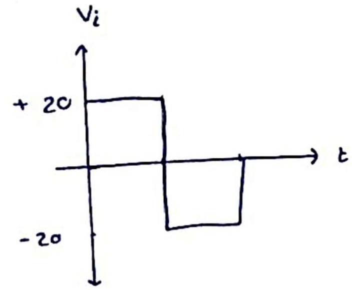
11. For a JFET, from bottom view, the pins X-Y-Z are:
a) Source - Gate - Drain
b) Gate - Source - Drain
c) Drain - Source - Gate
d) Gate - Drain - Source

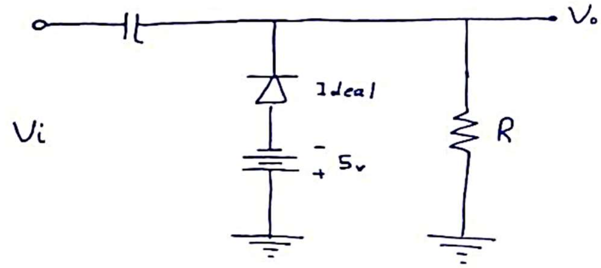


12. In the BJT experiment, the current gain/amplification for a common base is ………._, *and for a common emitter is............*
a) Ic / Ib , Ic / Ib
b) Ie / Ib , Ic / Ib
c) Ic / Ib , Ie / Ib
d) Ic / Ib , Ic / Ie

13. For the given clamber circuit, with ideal diode, the voltage of the capacitor is:

a) +15v
b) +14.3v
c) +20v
d) not mentioned





14. The FET transistor compared to the BJT is a……….. *with*…………..

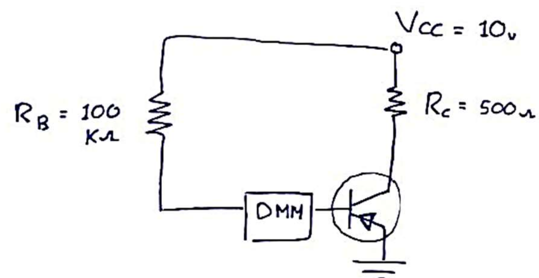
a) Voltage controlled device and a high input resistance
b) Voltage controlled device and a low input resistance
c) current controlled device and a low input resistance
d) current controlled device and a high input resistance

15. For a n-type FET characteristics curve is drawn for a ……….. values of V_{GS}

a) positive
b) negative
c) variable
d) not mentioned

16. For the given circuit, the BJT configuration is:__

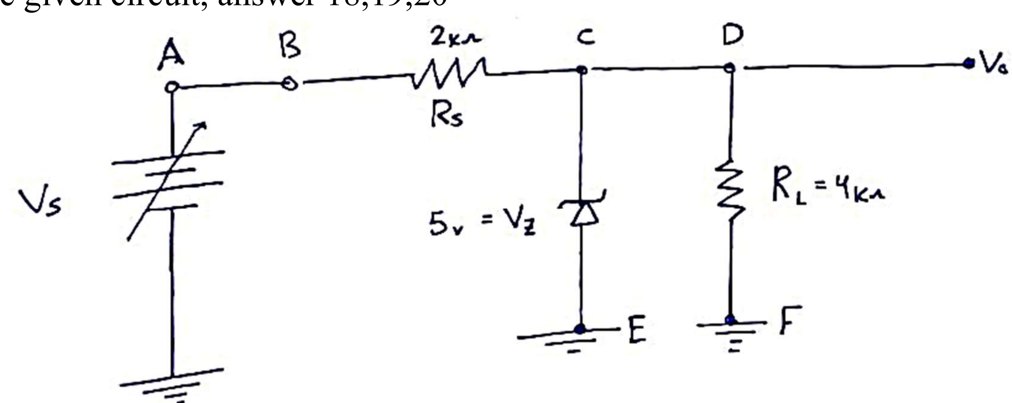
a) common base
b) common emitter
c) fixed biased
d) not mentioned



17.for the same circuit If the current gain was 100, and the current through resistor C is 9.3 mA then the reading of the DMM will be:

a) 93 μA
b) 68 μA
c) not determined
d) 0 mA

For the given circuit, answer 18,19,20



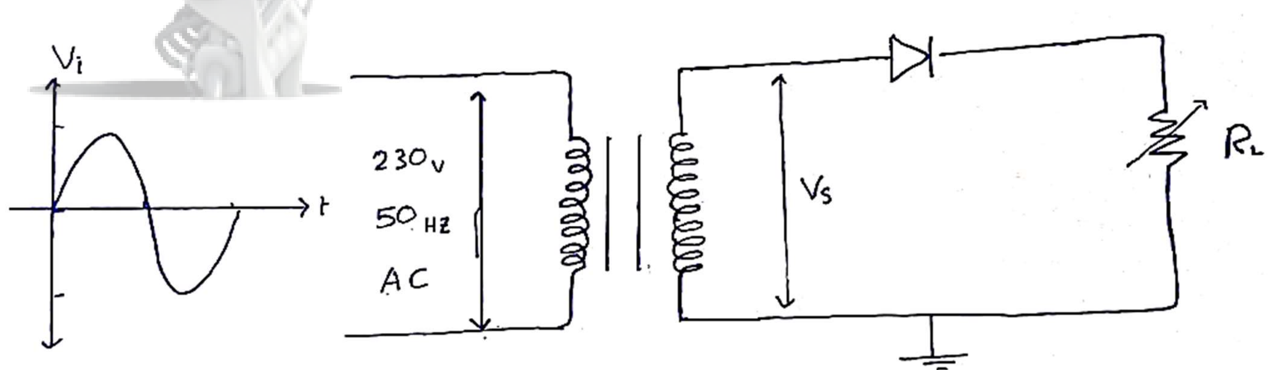
18. If Vs=4v, then the reading of the DMM between terminals A and B is:
   a) 0 mA
   b) 0.5 mA
   c) 1.3 mA
   d) not determined

19. If Vs=8v, then the reading of the DMM on the Rs is:
   a) 1.5 mA
   b) 3 v
   c) 13 v
   d) 5.5 v

20. If Vs=10v, then the reading of the DMM between terminals C and D is::
   a) 5v
   b) 1.25 v
   c) 2.5 mA
   d) not mentioned

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Question 2 : Given the circuit below, answer the following questions:



Q2.1) What is the name and objective of the given circuit?

Q2.2) What is the effect of adding a capacitor to the circuit?

Q2.3) Based on your answer in Q2.2, what is the effect of increasing the load resistor from 2 kΩ to 10 kΩ?

Q2.4) Explain how to measure the AC current through using only a Digital Multimeter (DMM).

Q2.5) Assume we connected an oscilloscope in parallel with R_L If the output frequency is 6 kHz, what would be the expected input frequency?

End.